EXPERIMENT 9

Aim: Implementation of Binary Search Technique considering a real-world application

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B2_24_E9.C

```
#include <stdio.h>
#include <conio.h>

void sort(int arr[100],int n)
{
int i,j,temp,c;
printf("enter your choice: \n");
scanf("%d",&c);
for(i=0;i<n-1;i++)
{
for(j=0;j<n-i-1;j++)
{
if(arr[j]>arr[j+1])
{
temp=arr[j];
arr[j]=arr[j+1];
arr[j+1]=temp;
}
}
}
}
```

1:76

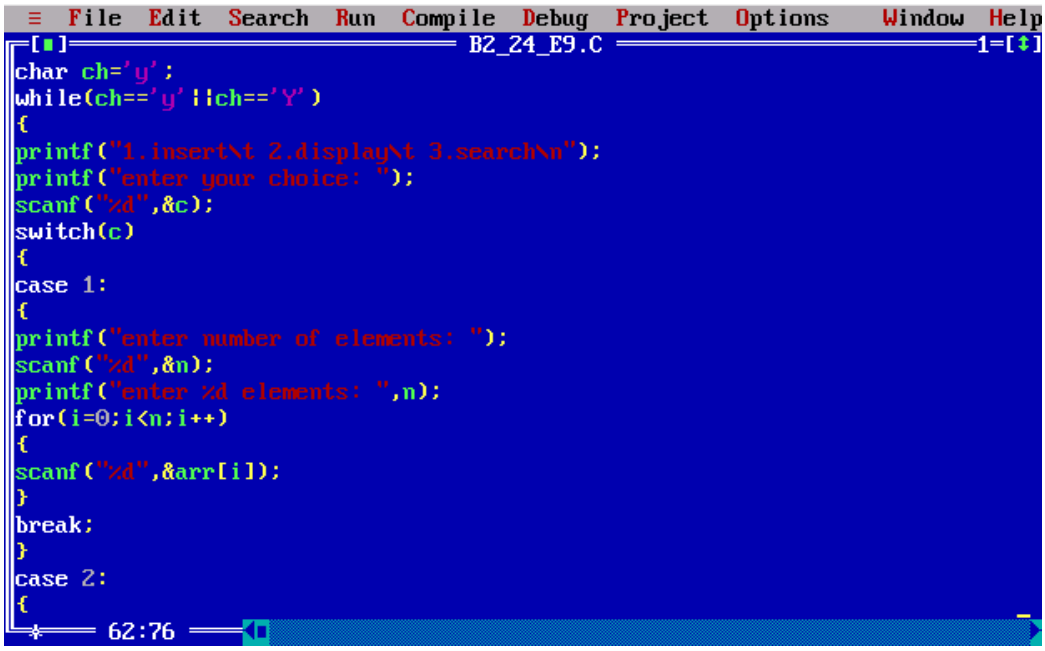
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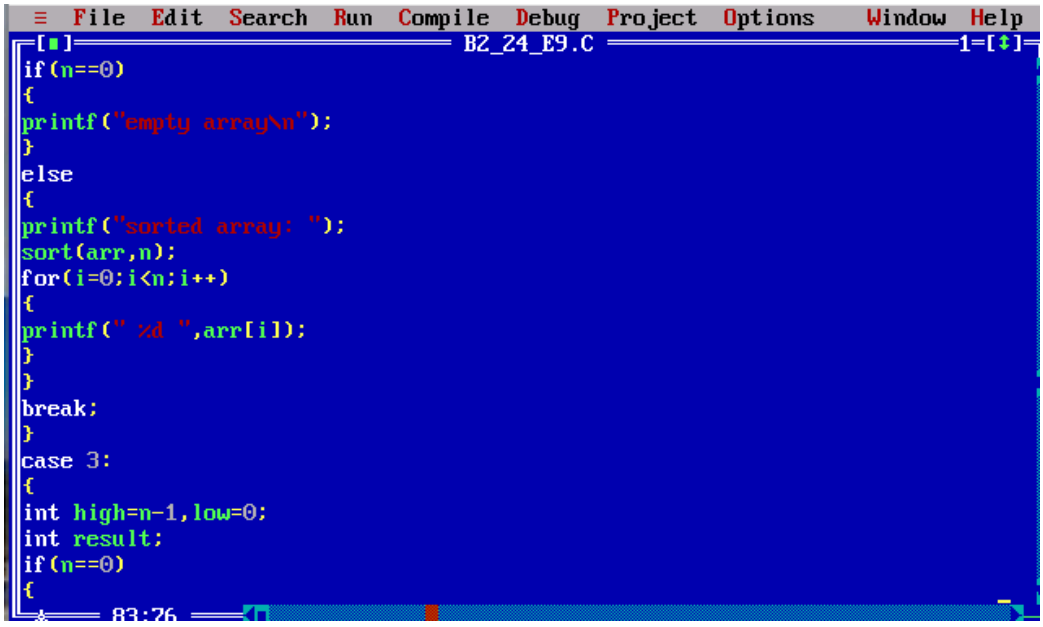
B2_24_E9.C

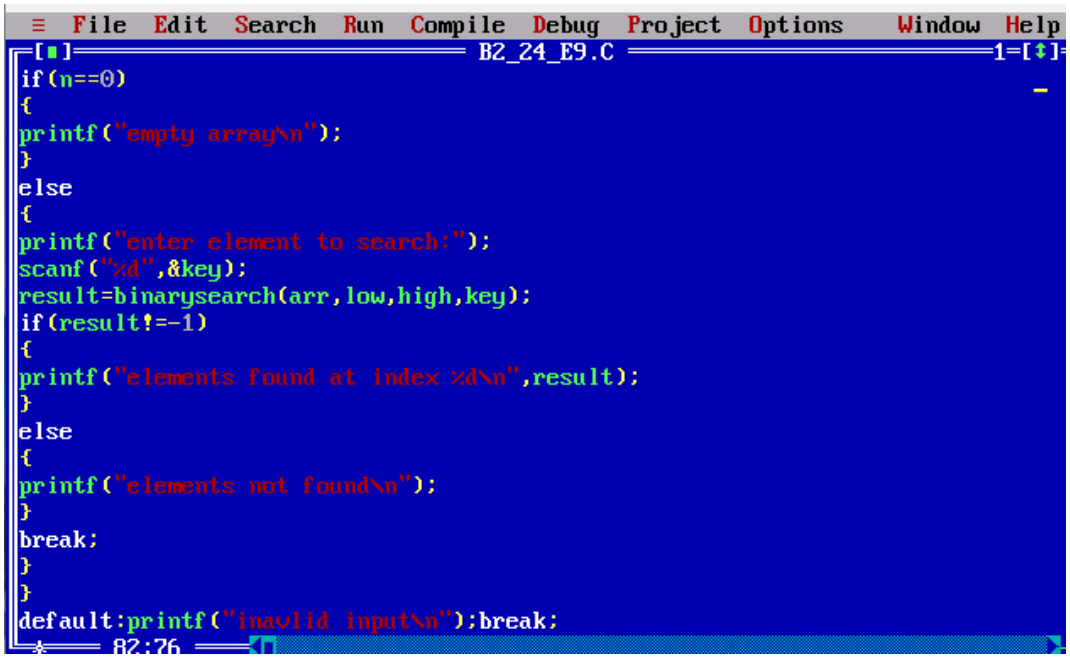
```
int binarysearch(int arr[],int low,int high,int key)
{
while(low<=high)
{
int mid=(low+high)/2;
if(arr[mid]==key)
return mid;
if(arr[mid]<key)
low=mid+1;
else
high=mid-1;
}
return -1;
}

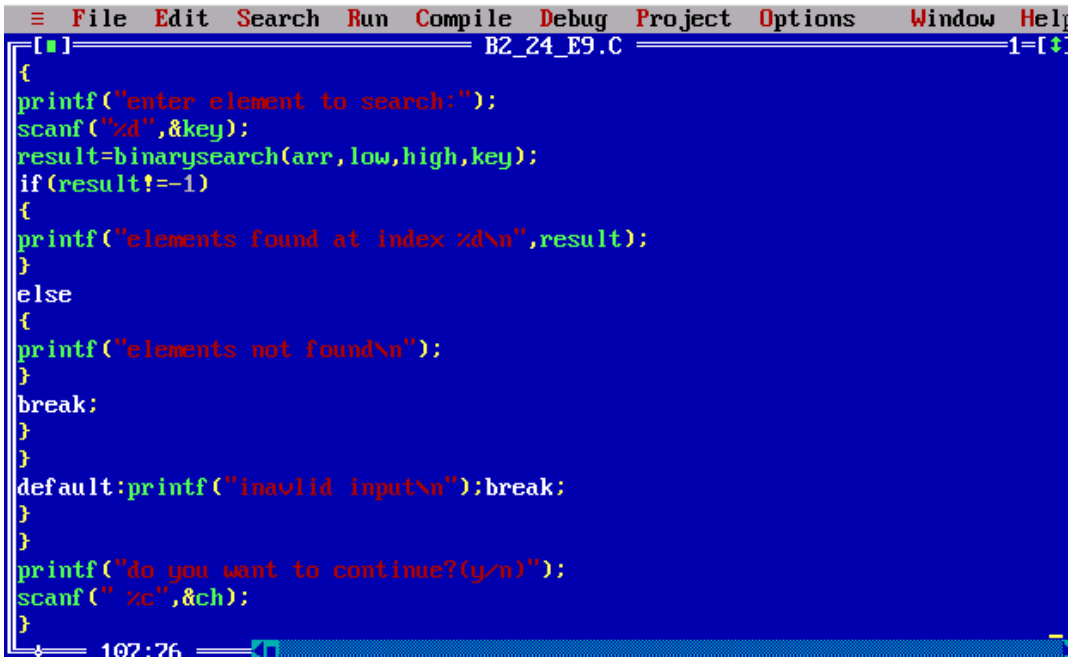
void main()
{
int i,n,c,key;
int arr[100];
char ch='y';
while(ch=='y'||ch=='Y')
```

43:76

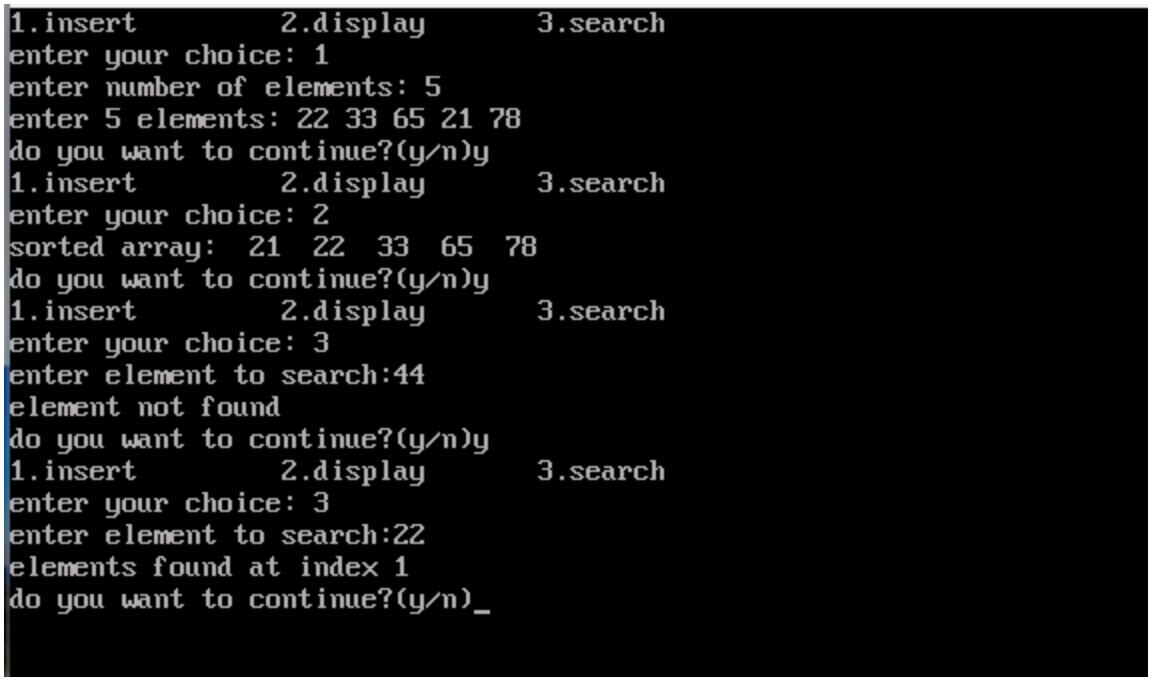

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B2_24_E9.C
1=[↕]
char ch='y';
while(ch=='y'||ch=='Y')
{
printf("1.insert\t 2.display\t 3.search\n");
printf("enter your choice: ");
scanf("%d",&c);
switch(c)
{
case 1:
{
printf("enter number of elements: ");
scanf("%d",&n);
printf("enter %d elements: ",n);
for(i=0;i<n;i++)
{
scanf("%d",&arr[i]);
}
break;
}
case 2:
{
62:76


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B2_24_E9.C
1=[↕]
if(n==0)
{
printf("empty array\n");
}
else
{
printf("sorted array: ");
sort(arr,n);
for(i=0;i<n;i++)
{
printf(" %d ",arr[i]);
}
}
break;
}
case 3:
{
int high=n-1,low=0;
int result;
if(n==0)
{
83:26


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B2_24_E9.C
if(n==0)
{
printf("empty array\n");
}
else
{
printf("enter element to search:");
scanf("%d",&key);
result=binarysearch(arr,low,high,key);
if(result!=-1)
{
printf("elements found at index %d\n",result);
}
else
{
printf("elements not found\n");
}
break;
}
}
default:printf("invalid input\n");break;
82:26


File Edit Search Run Compile Debug Project Options Window Help
B2_24_E9.C
{
printf("enter element to search:");
scanf("%d",&key);
result=binarysearch(arr,low,high,key);
if(result!=-1)
{
printf("elements found at index %d\n",result);
}
else
{
printf("elements not found\n");
}
break;
}
}
default:printf("invalid input\n");break;
}
}
printf("do you want to continue?(y/n)");
scanf(" %c",&ch);
}
102:26

Output:


1.insert 2.display 3.search
enter your choice: 1
enter number of elements: 5
enter 5 elements: 22 33 65 21 78
do you want to continue?(y/n)y
1.insert 2.display 3.search
enter your choice: 2
sorted array: 21 22 33 65 78
do you want to continue?(y/n)y
1.insert 2.display 3.search
enter your choice: 3
enter element to search:44
element not found
do you want to continue?(y/n)y
1.insert 2.display 3.search
enter your choice: 3
enter element to search:22
elements found at index 1
do you want to continue?(y/n)_